Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**4**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-3.66870618**
Energy E1	**-2.89838147**
Gap (E1-E0)	**0.770325**
<S^2>	**-0.000000**
S	**-0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

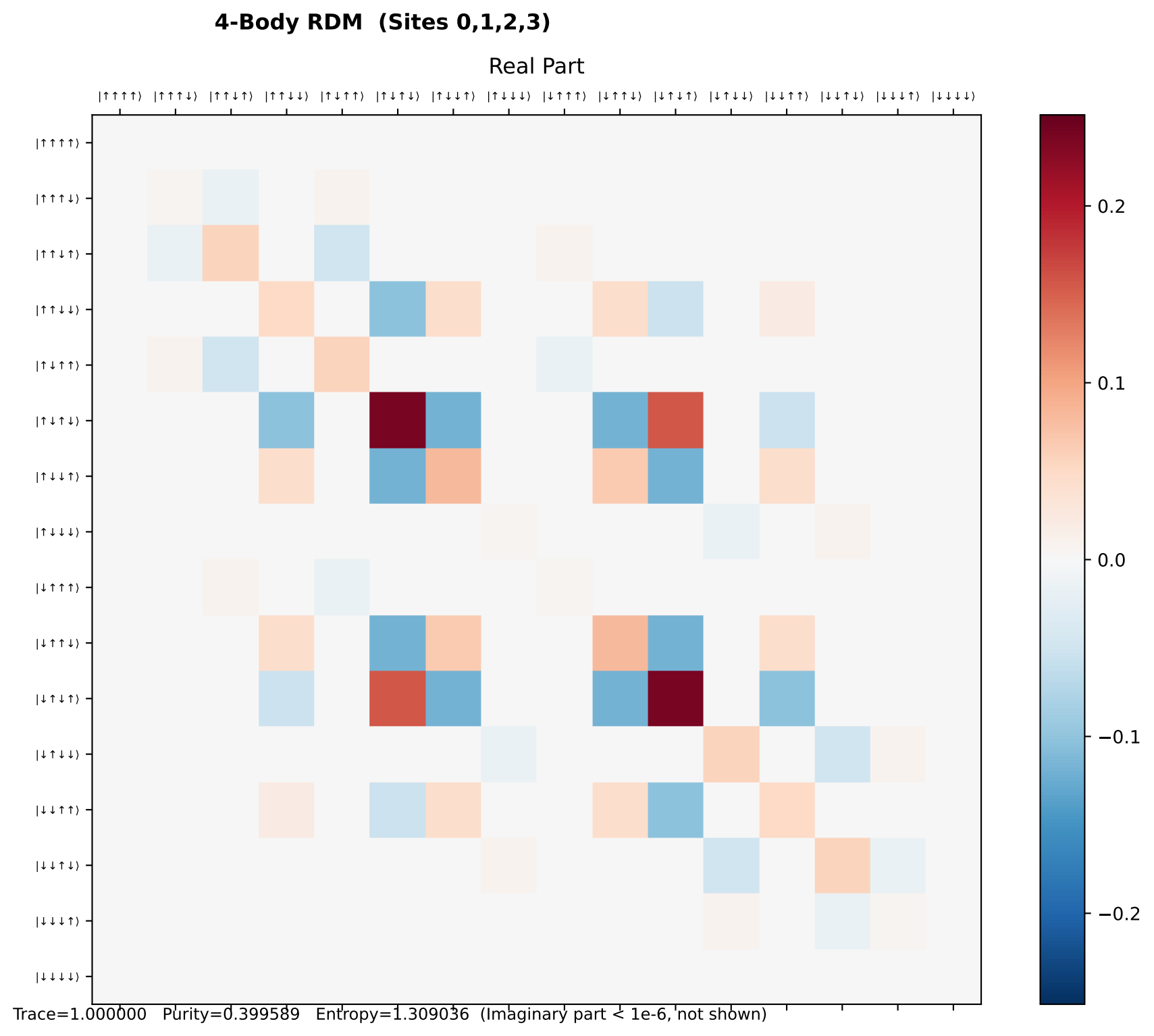
Double occupancy

<D> total	**0.666395**
<D> / site	**0.111066**

RDM and GME statistics for this run (4-body clusters)

Hilbert space dim : 2^4 = 16
Number of clusters : 6
Pure clusters : 0
Mixed clusters : 6
Clusters with GME (>0) : 6

Purity gamma = Tr(rho^2) : mean = 0.399589 [min 0.399589 , max 0.399589]
Von Neumann entropy : mean = 1.309036 [min 1.309036 , max 1.309036]
Rank of rho : mean = 11.00 [min 11 , max 11]
C_GME (best estimate) : mean = 0.233364 [min 0.218321 , max 0.241295]


4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -4.105692e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

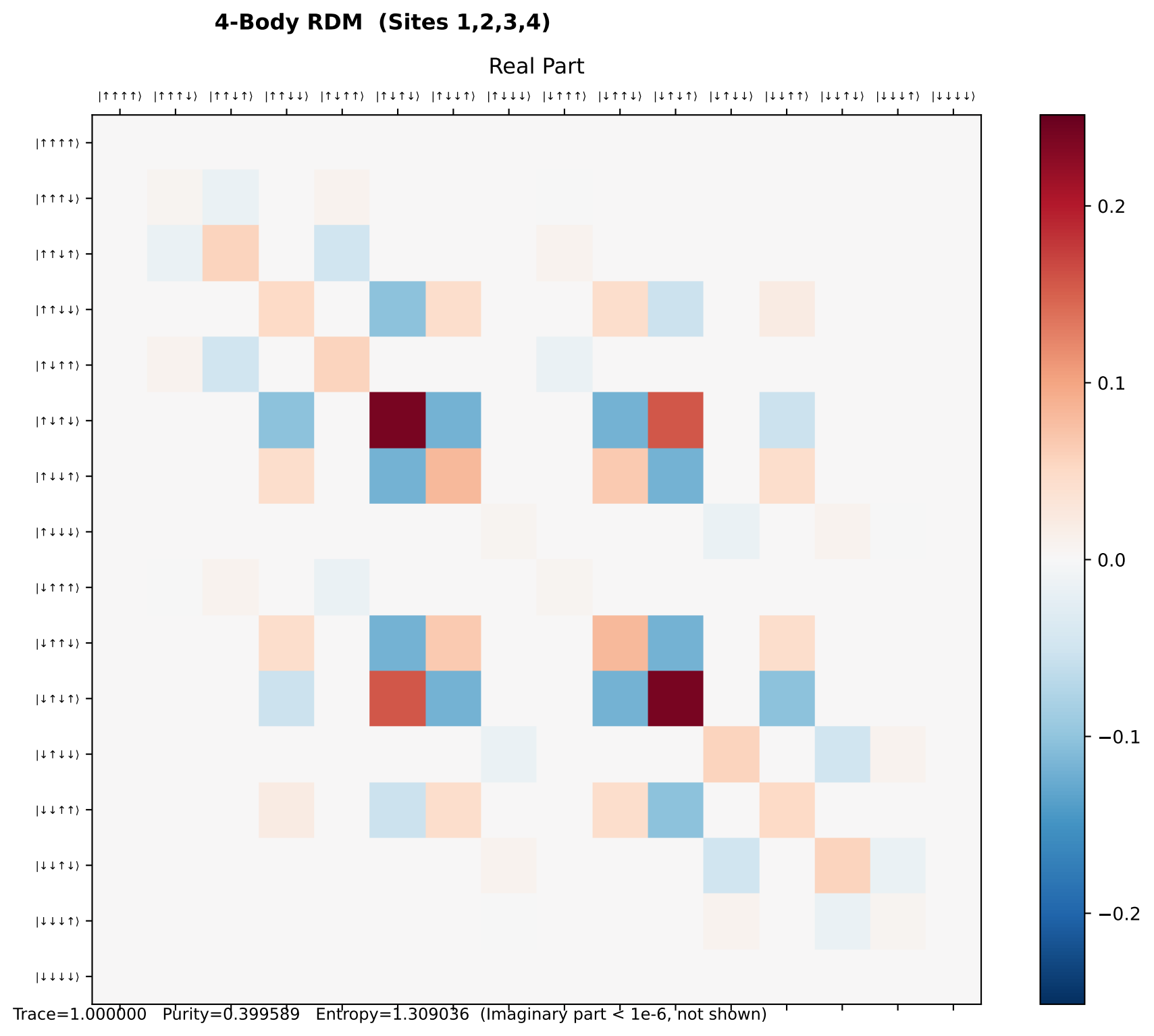
STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.23436667   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62716516

  Status: nONE
```


4-Body RDM (Sites 1,2,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,4)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -3.549109e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

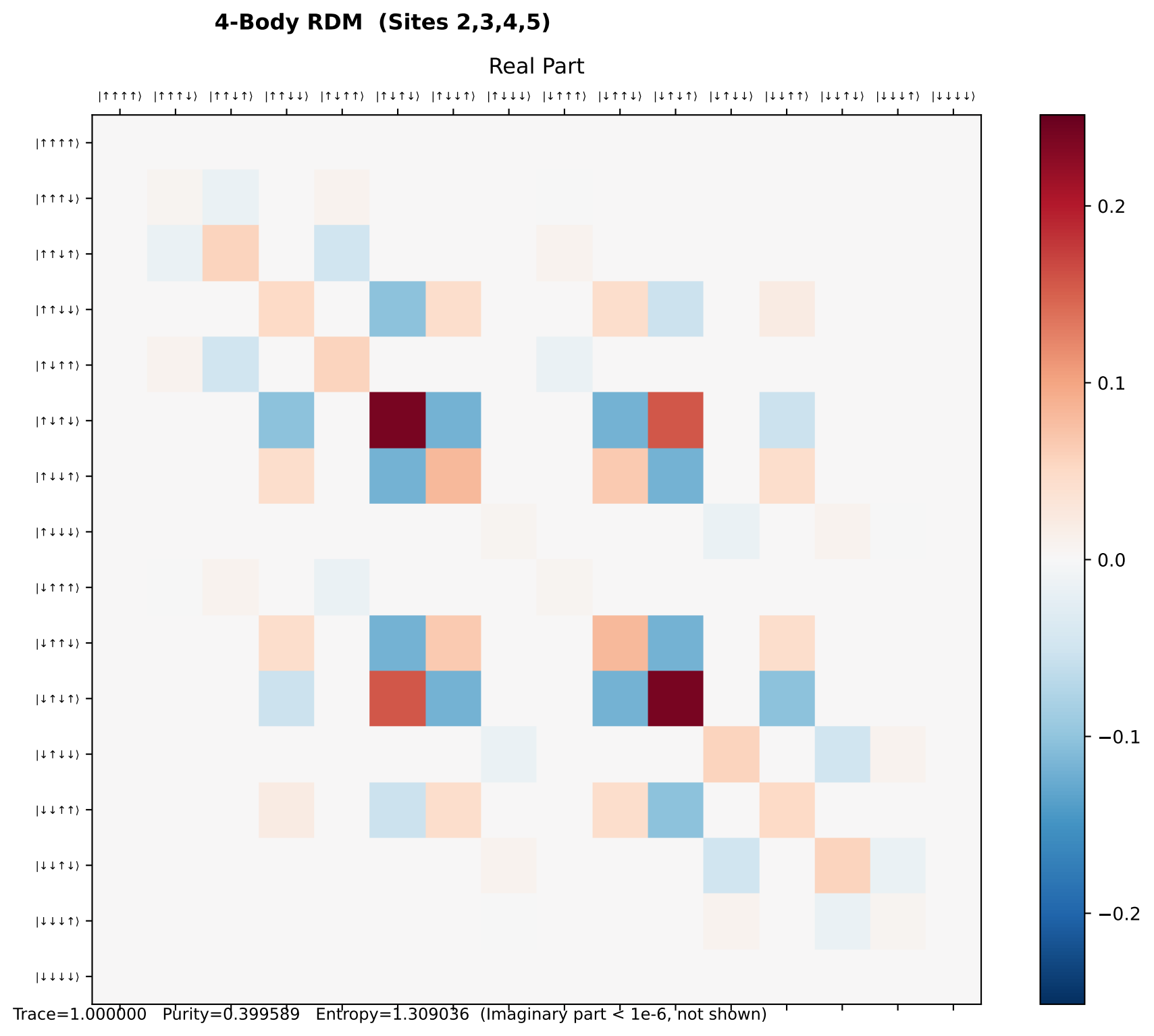
STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.21832117   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62765790

  Status: nONE
```


4-Body RDM (Sites 2,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 2,3,4,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : 0.000000e+00
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

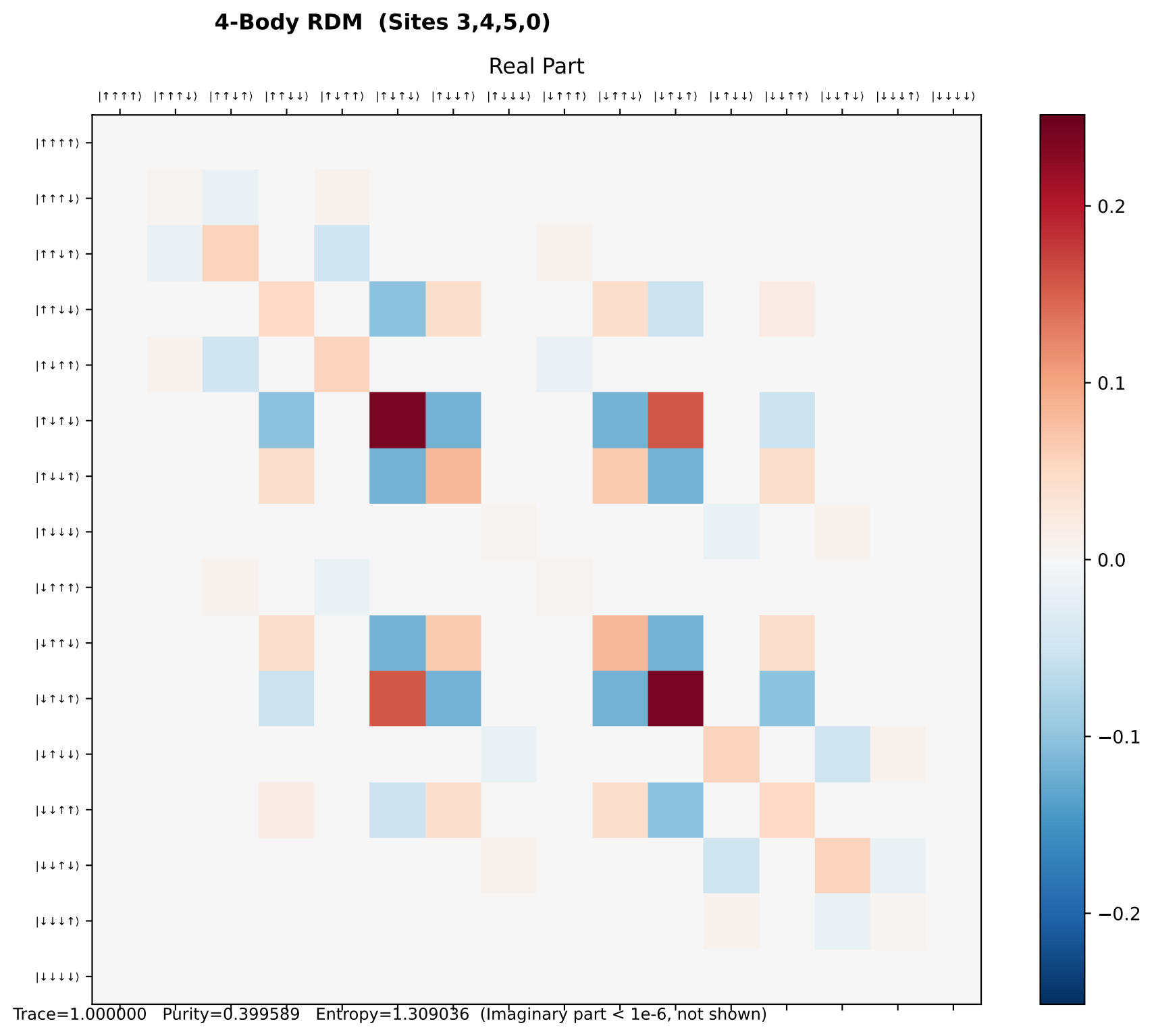
STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.24129453   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62686190

  Status: nONE
```


4-Body RDM (Sites 3,4,5,0)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 3,4,5,0)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -7.862403e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

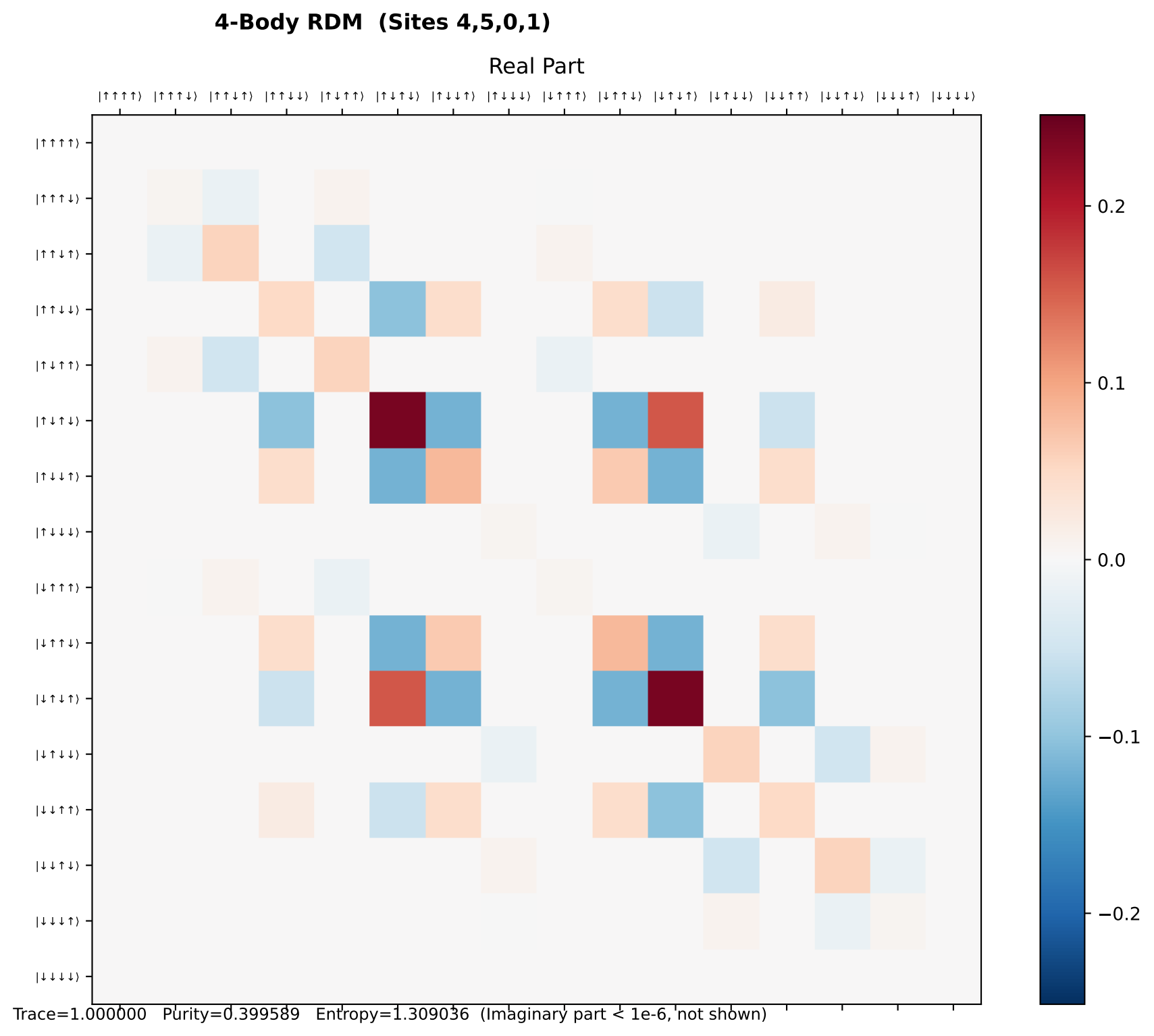
STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.23536242   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62695463

  Status: nONE
```


4-Body RDM (Sites 4,5,0,1)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 4,5,0,1)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -5.782549e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

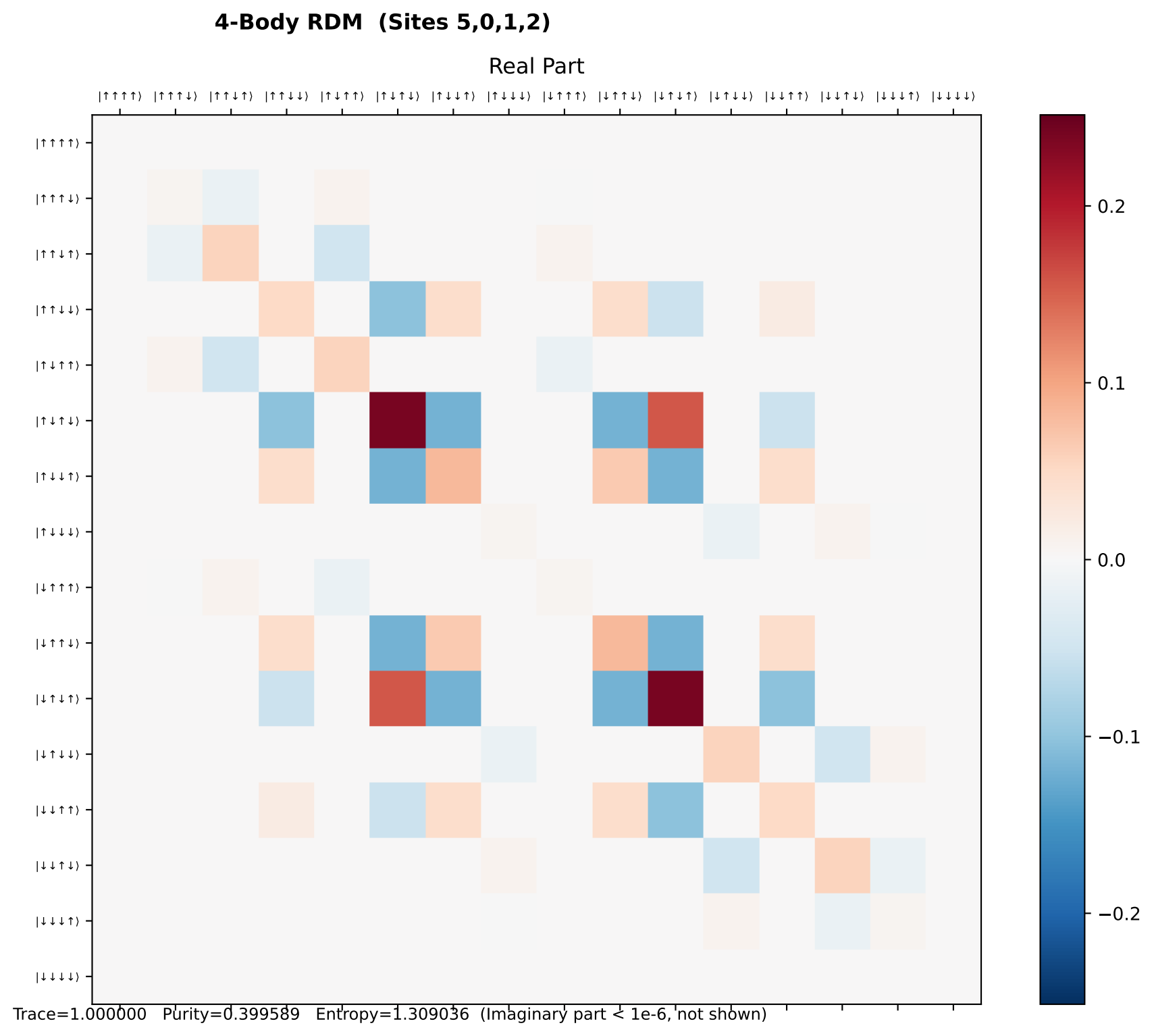
STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.23272350   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62763248

  Status: nONE
```


4-Body RDM (Sites 5,0,1,2)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.399589 Entropy=1.309036 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 5,0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3995892073
  Von Neumann entropy  : 1.3090357115
  Eigenvalue min       : -4.585787e-17
  Eigenvalue max       : 6.003318e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39958921

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.05288569
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.23811677   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.62765806

  Status: nONE
```